International Journal of Pharmaceutical, Engineering and Multidisciplinary Research (IJPEMR)

Article ID: IJPEMR-2026-002 | Volume 1, Issue 1 | 2026

Original Research Article

Extraction, Phytochemical Screening and In Vivo Anti -Diabetic Activity of Herbal Extract

Akash Kapse, Rahul Kumar Gupta

Radharaman College of Pharmacy, Bhopal 462044

Received: 27/07/2026 | Accepted: 02/08/2026 | Published: 16/08/2026

Study scope: Non-animal, non-human formulation development and in-vitro characterization

Abstract

Background: Diabetes mellitus is a metabolic disorder of rising global prevalence, and India carries one of the largest burdens of the disease. Medicinal plants are widely used in folk medicine as low-cost alternatives or adjuncts to synthetic antidiabetic drugs. *Luffa echinata* Roxb. (Cucurbitaceae) is traditionally used for several ailments, including diabetes.

Objective: To prepare a hydroalcoholic extract of *L. echinata* leaves, screen it for phytoconstituents, estimate its total phenolic and flavonoid content, and evaluate its antihyperglycemic activity in alloxan-induced diabetic rats.

Methods: Shade-dried leaves were defatted with petroleum ether and extracted by maceration with ethanol:water (70:20) for 48 h. Preliminary phytochemical screening was carried out by standard qualitative tests. Total phenolic content was estimated by the Folin-Ciocalteu method (gallic acid standard) and total flavonoid content by the aluminium chloride method (quercetin standard). Acute oral toxicity was assessed following OECD guideline 423. Diabetes was induced in Wistar rats with a single intraperitoneal dose of alloxan monohydrate (120 mg/kg). Diabetic rats received the extract at 100 and 200 mg/kg orally, or glibenclamide (600 µg/kg), for 15 days.

Results: [INSERT: phytochemical screening summary, e.g. which constituents were detected; TPC and TFC values in mg GAE/g and mg QE/g extract; percentage yield; fasting blood glucose at day 0 and day 15 for each group; body weight changes; lipid profile values; histopathology finding for pancreas.]

Conclusion: [INSERT after data is confirmed: e.g. The 200 mg/kg dose produced a significant antihyperglycemic effect compared with the 100 mg/kg dose and the diabetic control.] These findings support further investigation of *L. echinata* leaves as a source of antidiabetic agents, and the active constituents remain to be identified.

Keywords: *Luffa echinata*; antidiabetic activity; alloxan; hydroalcoholic extract; phytochemical screening; flavonoids; phenolics

1. Introduction

Diabetes mellitus is a group of heterogeneous metabolic disorders characterised by hyperglycaemia, which results from defects in insulin secretion, insulin action, or both (Sicree *et al.*, 2006). Chronic hyperglycaemia is associated with progressive damage to the eyes,

kidneys, nerves, heart and blood vessels, and it increases the risk of cardiovascular, peripheral vascular and cerebrovascular disease (WHO, 1999; Sicree *et al.*, 2006).

The disease is classified into type 1 diabetes, type 2 diabetes, gestational diabetes and other specific types (Sicree *et al.*, 2006). Type 2 diabetes accounts for roughly 85 to 95 percent of cases in high-income countries and an even larger share in developing countries (WHO, 1994). Its rising prevalence is linked to ageing populations, urbanisation, dietary change and reduced physical activity (WHO, 1994). The International Diabetes Federation estimated about 285 million adults with diabetes worldwide in 2010, a figure projected to reach 438 million by 2030 (Sicree *et al.*, 2006).

India has the largest number of people with diabetes of any country. The estimate for 2000 was about 31.7 million, and projections for 2030 approached 79 million (Wild *et al.*, 2004; Whiting *et al.*, 2011). Diabetes also imposes a heavy economic burden through treatment costs, disability, premature mortality and lost productivity (Kirigia *et al.*, 2009).

Current treatments include insulin, sulfonylureas, biguanides such as metformin, meglitinides, and lifestyle measures such as diet and exercise (Belinda, 2004; Votey and Peters, 2004). Plant-based medicines are attracting renewed interest because they are relatively inexpensive, widely available and, in many cases, have fewer side effects than synthetic drugs. India alone is estimated to have around 2,500 medicinal plant species, of which about 150 are used commercially on a large scale (Modak *et al.*, 2007). Numerous plants have been screened for antidiabetic activity in alloxan- and streptozotocin-induced rodent models (Venter *et al.*, 2008; Kar *et al.*, 2003).

Luffa echinata Roxb. is a spreading climber of the family Cucurbitaceae, found in Pakistan, India, Bangladesh and northern tropical Africa. In India it occurs mainly in Gujarat, Bihar, Rajasthan and Madhya Pradesh (Prakash *et al.*, 2013). The plant has a long history in traditional medicine for jaundice, colic, bronchitis, dropsy and diabetes (Modi and Kumar, 2014). Earlier studies have reported antiarthritic, hepatoprotective, antioxidant and hypoglycaemic activities of different parts of the plant (Chandel *et al.*, 2013; Ahmed *et al.*, 2001; Kumar *et al.*, 2000). The hydroalcoholic extract has also been reported to produce hypoglycaemic action in rats (Sharma *et al.*, 2012).

Despite these reports, the antidiabetic potential of the leaves has not been studied in detail. The present study therefore aimed to prepare a hydroalcoholic extract of *L. echinata* leaves, screen

it for phytoconstituents, quantify its phenolic and flavonoid content, and evaluate its antihyperglycaemic activity in alloxan-induced diabetic rats.

2. Review of Literature

2.1 *Luffa echinata*

Modi and Kumar (2014) reviewed the ethnomedicinal, phytochemical and pharmacological properties of *L. echinata*. The plant has been traditionally used against jaundice, intestinal colic, enlarged liver and spleen, leprosy, diabetes, bronchitis, nephritis, rheumatism and fever. Reported pharmacological activities include antioxidant, analgesic, anti-inflammatory, antidepressant, anxiolytic, antiepileptic, hepatoprotective, antibacterial, antifungal, antiulcer and anticancer effects.

Chandel *et al.* (2013) studied the aqueous and alcoholic extracts of *L. echinata* fruit in Freund's complete adjuvant-induced arthritis in rats. Both extracts reduced paw oedema and restored haematological parameters, with the effect at 250 mg/kg reported as comparable across the developing and developed phases of arthritis. Ahmed *et al.* (2001) evaluated fruit extracts against CCl_4-induced hepatotoxicity and reported that petroleum ether, acetone and methanolic extracts gave hepatoprotection comparable to silymarin.

2.2 Plants Investigated for Antidiabetic Activity

A large number of plants have been tested for antidiabetic activity in experimental animals. Studies on *Caralluma sinaica* (Habibuddin *et al.*, 2008), *Phyllanthus reticulatus* (Kumar *et al.*, 2008), *Murraya koenigii* (Kesari *et al.*, 2006; Yadav *et al.*, 2002), *Caesalpinia bonducella* (Kannur *et al.*, 2006), *Aegle marmelos* (Kesari *et al.*, 2006) and *Catharanthus roseus* (Srinivas *et al.*, 2003; Singh *et al.*, 2001) reported dose-dependent reductions in blood glucose. Eidi *et al.* (2006) showed that garlic ethanolic extract lowered serum glucose, lipids and transaminases in streptozotocin-induced rats. Chattopadhyay (1999) compared four medicinal plants and found *Azadirachta indica* leaf extract to have the most potent blood-sugar-lowering effect, followed by *C. roseus*, *Gymnema sylvestre* and *Ocimum sanctum*. Several of these studies attributed the effect partly to antioxidant activity, which supports testing phenolic and flavonoid content alongside glucose-lowering activity (Raut and Gaikwad, 2006; Kamalakkannan and Prince, 2006).

3. Research Envisaged

Diabetes mellitus can be managed with diet, exercise and drugs, but synthetic agents are not free of adverse effects, and plant extracts remain an important source of alternatives (Votey and Peters, 2004). The aim of this study was to assess the extraction, phytochemical screening and in vivo antidiabetic activity of *L. echinata* leaf extract.

The specific objectives were:

1. To prepare a hydroalcoholic extract of *L. echinata* leaves and determine its percentage yield.
2. To perform qualitative phytochemical screening for major phytoconstituents.
3. To estimate total phenolic and total flavonoid content.
4. To assess acute oral toxicity of the extract in rats.
5. To evaluate the antihyperglycaemic activity of the extract at two doses in alloxan-induced diabetic rats, using glibenclamide as the reference drug.

4. Materials and Methods

4.1 Plant Material

Fresh leaves of *L. echinata* were collected from [INSERT: exact location, district, state] in [INSERT: month, year]. [INSERT: botanical identification by name, institution and voucher specimen number.] The plant was selected on the basis of its local availability and its use in folk medicine.

The leaves were shade-dried at room temperature, coarsely powdered, packed in airtight polythene bags and stored until use.

4.2 Extraction

The shade-dried powder was first defatted with petroleum ether by maceration until defatting was complete (Mukherjee, 2007; Kokate, 1994). The defatted material was then extracted with hydroalcoholic solvent (ethanol:water, 70:20 [INSERT: confirm the complete ratio, e.g. 70:30]) by maceration for 48 h. The filtrate was concentrated under vacuum at 40 °C.

The percentage yield was calculated as:

Percentage yield = (weight of extract ÷ weight of powdered drug taken) × 100

[INSERT: weight of powder taken, weight of extract obtained, and percentage yield.]

4.3 Preliminary Phytochemical Screening

Qualitative tests were carried out on the extract using standard methods (Kokate, 1994; Roopashree *et al.*, 2008). Table 1 lists the tests applied.

Table 1. Qualitative phytochemical tests performed on the extract

Constituent	Test	Positive result
Alkaloids	Mayer's, Wagner's, Dragendorff's, Hager's	Yellow, brown/reddish, red or yellow precipitate
Carbohydrates	Molisch's, Benedict's, Fehling's	Violet ring; orange-red or red precipitate
Glycosides	Modified Borntrager's, Legal's	Rose-pink (anthranol); pink to blood-red (cardiac)
Saponins	Froth test, foam test	1 cm stable foam layer; foam persisting 10 min
Phytosterols / triterpenes	Salkowski's, Liebermann-Burchard	Golden yellow (triterpenes); brown ring (phytosterols)
Phenols	Ferric chloride	Bluish-black colour
Tannins	Gelatin test	White precipitate
Flavonoids	Alkaline reagent, lead acetate	Intense yellow (turns colourless with dilute acid); yellow precipitate
Proteins and amino acids	Xanthoproteic, ninhydrin	Yellow colour; blue colour
Diterpenes	Copper acetate	Emerald green colour

[INSERT: observed result for each test, e.g. "+" for present and "–" for absent.]

4.4 Estimation of Total Phenolic Content

The total phenolic content was determined by a modified Folin-Ciocalteu method (Olajuyigbe and Afolayan, 2011). A gallic acid stock solution (50 mg in 50 ml methanol) was diluted to obtain standards in the range of 5 to 25 µg/ml. The extract (10 mg in 10 ml methanol, filtered; 1 mg/ml) was prepared in the same way. For the assay, 2 ml of each standard or extract was mixed with 1 ml of Folin-Ciocalteu reagent (diluted 1:10 with distilled water) and 1 ml of sodium carbonate (7.5 g/l). The mixture was vortexed for 15 s and kept for 15 min for colour development. Absorbance was read at 765 nm. A calibration curve was plotted, and the results were expressed as mg gallic acid equivalent (GAE) per gram of extract.

4.5 Estimation of Total Flavonoid Content

Total flavonoids were estimated by the aluminium chloride colourimetric method (Olajuyigbe and Afolayan, 2011). A quercetin stock (50 mg in 50 ml methanol) was diluted to 5 to 25 µg/ml. For the assay, 1 ml of 2% $AlCl_3$ solution was added to 3 ml of standard or extract solution, and

the mixture was left at room temperature for 15 min. Absorbance was measured at 420 nm, and results were expressed as mg quercetin equivalent (QE) per gram of extract.

4.6 Animals

Wistar rats weighing 150 to 200 g were housed in groups of six under a 12 h light/dark cycle at 25 ± 2 °C, with standard rodent chow and water ad libitum. Animals were acclimatised for 7 days before the experiments. All procedures were carried out between 08:00 and 15:00 h in a noise-free room. Each experiment used a separate group of six rats. The study was approved by the Institutional Animal Ethics Committee (IAEC) [INSERT: approval number and date].

4.7 Chemicals

Alloxan monohydrate (Central Drug House Pvt. Ltd., India) and glibenclamide tablets (Daonil; Aventis Pharma Ltd., India) were obtained from authorised distributors. All other chemicals were of analytical grade. [INSERT: confirm all other chemical suppliers.]

4.8 Acute Oral Toxicity

Acute oral toxicity was assessed following OECD guideline 423 (OECD, 2002). The extract was administered orally at 50, 100, 150, 200 and 300 mg/kg to groups of rats, which were observed for mortality and behavioural changes. [INSERT: number of animals per dose, observation period, and outcome. Note: the original protocol lists six groups but five doses.]

4.9 Induction of Diabetes

Rats were fasted overnight and then given a single intraperitoneal injection of alloxan monohydrate (120 mg/kg body weight) in distilled water. To prevent alloxan-induced hypoglycaemia, the animals were given 5% glucose solution to drink overnight (Adediwura and Abo Kio, 2008). Fasting blood glucose was measured 15 days after induction, and animals with elevated fasting glucose were selected for the study (Pareek *et al.*, 2009). [INSERT: threshold glucose value used to define diabetes.]

4.10 Experimental Design

Animals were divided into five groups of six rats each (Pareek *et al.*, 2009):

- **Group I (normal control):** received vehicle (0.5 ml distilled water/rat/day).
- **Group II (diabetic control):** received vehicle (0.5 ml distilled water/rat/day).
- **Group III (diabetic + extract):** received hydroalcoholic extract of *L. echinata* at 100 mg/kg p.o. for 15 days.

- **Group IV (diabetic + extract):** received hydroalcoholic extract of *L. echinata* at 200 mg/kg p.o. for 15 days.
- **Group V (diabetic + reference):** received glibenclamide at 600 µg/kg p.o. for 15 days.

Fasting blood glucose, body weight and [INSERT: other parameters, e.g. serum lipid profile] were recorded at [INSERT: days of measurement]. At the end of the study, [INSERT: method of sacrifice and tissue collection], and the pancreas was examined histologically.

4.11 Statistical Analysis

[INSERT: statistical test used, e.g. one-way ANOVA followed by Dunnett's or Tukey's post hoc test; the significance level; the software used.]

5. Results

5.1 Extraction Yield and Phytochemical Screening

[INSERT: percentage yield of the extract, followed by a sentence summarising which constituents were detected. Refer to Table 1 and include the result table.]

5.2 Phenolic and Flavonoid Content

The total phenolic content was [INSERT: value] mg GAE/g of extract, and the total flavonoid content was [INSERT: value] mg QE/g of extract (Table 2).

Table 2. Total phenolic and flavonoid content of the hydroalcoholic extract

Parameter	Mean ± SD (n = [INSERT])
Total phenolic content (mg GAE/g extract)	[ ]
Total flavonoid content (mg QE/g extract)	[ ]

5.3 Acute Oral Toxicity

[INSERT: mortality and behavioural observations at each dose. State whether the LD50 cut-off was exceeded.]

5.4 Effect on Fasting Blood Glucose

Table 3 summarises the fasting blood glucose levels in each group before and after 15 days of treatment.

Table 3. Effect of *L. echinata* extract on fasting blood glucose (mg/dl) in alloxan-induced diabetic rats

Group	Day 0	Day 15	% change
I: Normal control	[ ]	[ ]	[ ]
II: Diabetic control	[ ]	[ ]	[ ]
III: Extract 100 mg/kg	[ ]	[ ]	[ ]
IV: Extract 200 mg/kg	[ ]	[ ]	[ ]
V: Glibenclamide 600 µg/kg	[ ]	[ ]	[ ]

Values are mean ± SD, n = 6. [INSERT: significance markers, e.g. * $p < 0.05$ vs diabetic control.]

5.5 Effect on Body Weight and Lipid Profile

[INSERT: body weight changes and lipid profile values (total cholesterol, triglycerides, HDL, LDL) for each group, with a table.]

5.6 Histopathology of the Pancreas

[INSERT: description of pancreatic islet morphology in each group, with a representative figure.]

6. Discussion

The present study prepared a hydroalcoholic extract of *L. echinata* leaves and evaluated it for phytochemical content and antihyperglycaemic activity in alloxan-induced diabetic rats. Alloxan selectively destroys pancreatic beta cells, which leads to insulin deficiency and sustained hyperglycaemia (Szkudelski, 2012). The model is therefore suitable for screening agents that lower blood glucose through insulin-dependent or insulin-independent mechanisms.

[INSERT: Interpret the phytochemical results. Flavonoids and phenolics are commonly associated with antioxidant and glucose-lowering effects (Raut and Gaikwad, 2006; Kamalakkannan and Prince, 2006). Compare the TPC and TFC values with those reported for other plants if possible.]

[INSERT: Compare the glucose-lowering effect of the 200 mg/kg dose with the 100 mg/kg dose, and with glibenclamide. Discuss whether the dose-response relationship is consistent with earlier reports of dose-dependent activity in *L. echinata* and other plants (Kumar *et al.*, 2008; Ndiaye *et al.*, 2008).]

[INSERT: Discuss the body weight, lipid profile and pancreatic histopathology findings. Improvement in lipid profile is relevant because diabetes is associated with dyslipidaemia and cardiovascular risk (Collins, 2002; Shepherd, 2005).]

[INSERT: Limitations. For example: a single alloxan dose, a short 15-day treatment period, no mechanistic study, and the lack of isolated active compounds. Also note that the study used the leaves, while some earlier work used fruits or roots.]

7. Conclusion

[INSERT, based only on confirmed data: The hydroalcoholic extract of *L. echinata* leaves contains phenolic and flavonoid constituents and showed [describe effect] on fasting blood glucose in alloxan-induced diabetic rats. The 200 mg/kg dose was more effective than the 100 mg/kg dose in lowering blood glucose, and it also improved [body weight / lipid profile / pancreatic histology].] Further studies are needed to identify the active constituent(s) responsible for the antidiabetic effect and to confirm the findings with longer treatment and clinical data.

Acknowledgements

[INSERT]

References

1. Ahmed, B., Alam, T., and Khan, S. A. (2001). Hepatoprotective activity of *Luffa echinata* fruits. *Journal of Ethnopharmacology*, 76, 187–189.
2. Adediwura, F.-J., and Kio, A. (2008). Antidiabetic activity of *Gladiolus psittacinus* in alloxan-induced diabetic rats. *African Journal of Traditional, Complementary and Alternative Medicines*, 5(2), 154–157.
3. Chattopadhyay, R. R. (1999). A comparative evaluation of some blood sugar lowering agents of plant origin. *Journal of Ethnopharmacology*, 67(3), 367–372.
4. Collins, F. M. (2002). Current treatment approaches to type 2 diabetes mellitus: successes and shortcomings. *American Journal of Managed Care*, 8(16 Suppl), S460–S471.

5. Eidi, A., Eidi, M., and Esmaeili, E. (2006). Antidiabetic effect of garlic (*Allium sativum* L.) in normal and streptozotocin-induced diabetic rats. *Phytomedicine*, 13(9–10), 624–629.

6. Habibuddin, M., Daghriri, H. A., Humaira, T., and Hefzi, A. A. H. (2008). Antidiabetic effect of alcoholic extract of *Caralluma sinaica* L. on streptozotocin-induced diabetic rabbits. *Journal of Ethnopharmacology*, 117(2), 215–220.

7. Kamalakkannan, N., and Prince, P. S. (2006). Antihyperglycaemic and antioxidant effect of rutin, a polyphenolic flavonoid, in streptozotocin-induced diabetic Wistar rats. *Basic and Clinical Pharmacology and Toxicology*, 98, 97–103.

8. Kannur, D. M., Hukkeri, V. I., and Akki, K. S. (2006). Antidiabetic activity of *Caesalpinia bonducella* seed extracts in rats. *Fitoterapia*, 77(7–8), 546–549.

9. Kar, A., Choudhary, B. K., and Bandyopadhyay, N. G. (2003). Comparative evaluation of hypoglycaemic activity of some Indian medicinal plants in alloxan diabetic rats. *Journal of Ethnopharmacology*, 84, 105–108.

10. Kesari, A. N., Gupta, R. K., Singh, S. K., Diwakar, S., and Watal, G. (2006). Hypoglycemic and antihyperglycemic activity of *Aegle marmelos* seed extract in normal and diabetic rats. *Journal of Ethnopharmacology*, 107(3), 374–379.

11. Kesari, A. N., Gupta, R. K., and Watal, G. (2005). Hypoglycemic effects of *Murraya koenigii* on normal and alloxan-diabetic rabbits. *Journal of Ethnopharmacology*, 97(2), 247–251.

12. Kirigia, J. M., Sambo, H. B., Sambo, L. G., and Barry, S. P. (2009). Economic burden of diabetes mellitus in the WHO African region. *BMC International Health and Human Rights*, 9, 6.

13. Kokate, C. K. (1994). *Practical Pharmacognosy* (4th ed.). Vallabh Prakashan, pp. 112–120.

14. Kumar, S., Kumar, D., Deshmukh, R. R., Lokhande, P. D., More, S. N., and Rangari, V. D. (2008). Antidiabetic potential of *Phyllanthus reticulatus* in alloxan-induced diabetic mice. *Fitoterapia*, 79(1), 21–23.

15. Kumar, V. P., Shashidhara, S., Kumar, M. M., and Sridhara, B. Y. (2000). Effect of *Luffa echinata* on lipid peroxidation and free radical scavenging activity. *Journal of Pharmacy and Pharmacology*, 52(7), 891–894.

16. Mukherjee, P. K. (2007). *Quality Control of Herbal Drugs* (2nd ed.). Business Horizons, pp. 2–14.

17. Modak, M., Dixit, P., Londhe, J., and Devasagayam, T. P. A. (2007). Indian herbs and herbal drugs used for the treatment of diabetes. *Journal of Clinical Biochemistry and Nutrition*, 40(3), 163–173.

18. Modi, A., and Kumar, V. (2014). *Luffa echinata* Roxb.: A review on its ethnomedicinal, phytochemical and pharmacological perspective. *Asian Pacific Journal of Tropical Disease*, 4(1), 7–12.

19. Ndiaye, M., Diatta, W., Sy, A. N., Dièye, A. M., Faye, B., and Bassène, E. (2008). Antidiabetic properties of aqueous barks extract of *Parinari excelsa* in alloxan-induced diabetic rats. *Fitoterapia*, 79, 267–270.

20. OECD. (2002). *Acute oral toxicity: Acute toxic class method, Guideline 423*. Organisation for Economic Co-operation and Development, Paris.

21. Olajuyigbe, O. O., and Afolayan, A. J. (2011). Phenolic content and antioxidant property of the bark extract of *Ziziphus mucronata* Willd. subsp. *mucronata*. *BMC Complementary and Alternative Medicine*, 11, 130.

22. Pareek, H., Sharma, S., Khajja, B. S., Jain, K., and Jain, G. C. (2009). Evaluation of hypoglycemic and anti-hyperglycemic potential of *Tridax procumbens* (Linn.). *BMC Complementary and Alternative Medicine*, 9, 48.

23. Prakash, K., Pandey, A., Radhamani, J., and Bisht, I. S. (2013). Morphological variability in cultivated and wild species of *Luffa* (Cucurbitaceae) from India. *Genetic Resources and Crop Evolution*, 60(8), 2319–2329.

24. Raut, N. A., and Gaikwad, N. J. (2006). Antidiabetic activity of hydro-ethanolic extract of *Cyperus rotundus* in alloxan-induced diabetes in rats. *Fitoterapia*, 77(7–8), 585–588.

25. Roopashree, T. S., Dang, R., Rani, S. R. H., and Narendra, C. (2008). Antibacterial activity of anti-psoriatic herbs: *Cassia tora*, *Momordica charantia* and *Calendula officinalis*. *International Journal of Applied Research in Natural Products*, 1(3), 20–28.

26. Sharma, T., Arora, R., and Gill, N. S. (2012). Evaluation of free radical scavenging, anti-inflammatory and analgesic potential of *Luffa echinata* seed extract. *Journal of Medical Sciences*, 12(4), 99–106.

27. Shepherd, J. (2005). Does statin monotherapy address the multiple lipid abnormalities in type-2 diabetes? *Atherosclerosis Supplements*, 6, 15–19.

28. Sicree, R., Shaw, J., and Zimmet, P. (2006). The global burden. Diabetes and impaired glucose tolerance: Prevalence and projections. In D. Gan (Ed.), *Diabetes Atlas* (3rd ed., pp. 16–103). International Diabetes Federation.

29. Singh, S. N., Vats, P., Suri, S., and Sridharan, K. (2001). Effect of an antidiabetic extract of *Catharanthus roseus* on enzymatic activities in streptozotocin-induced diabetic rats. *Journal of Ethnopharmacology*, 76(3), 269–277.

30. Chandel, H. S., Singh, S., and Kushwaha, R. (2013). Evaluation of antiarthritic activity on *Luffa echinata* Roxb. fruits on rats. *Asian Journal of Biomedical and Pharmaceutical Sciences*, 3(21), 36–41.